

SUMMARY

Applied scientist and research engineer with 6+ years of experience taking multimodal models from data design and pre/post-training through rigorous evaluation and production deployment. Led end-to-end VLM and visual-grounding projects and teams of up to eight people, built scalable PyTorch training/serving systems, and contributed 40+ pull requests to Hugging Face Transformers.

EXPERIENCE

- **Toss Bank** Seoul, South Korea
Data Scientist *Jan 2026 - Present*
 - **On-premise AI Agent System:** Building on-premise tool-calling agents that route requests between direct question answering and RAG over internal documents retrieved with image embeddings.
 - **AI Authentication – My ID MFR:** Developing My ID MFR for mobile ID card recognition and facial verification with ONNX models running through WebGPU; combining face anti-spoofing, multitask depth/normal/keypoint estimation, and face-embedding similarity.
 - **Language Model Evaluation:** Building an in-house language model benchmark aligned with public evaluation protocols and scoring heuristics to compare Qwen and GLM models and analyze performance differences.
 - **[Lead] End-to-End Document Information Extraction:** Led the full research and production pipeline for an approximately 1B-parameter vision-language model spanning multi-page classification, orientation, layout detection, OCR/table understanding, and LLM-based key-value extraction; post-trained the model, designed stage-level diagnostics, and reached 61% exact-match accuracy for automation-ready outputs.
- **SuperbAI** Seoul, South Korea
Machine Learning Engineer *Sep 2021 - Jan 2026*
 - **[Lead] Multimodal Foundation Model:** Held approximately one year of team-leadership responsibility, leading two ML engineers through multimodal pre-training and staged text/image alignment for a visual-grounding model, from planning and dataset construction to evaluation and deployment.
 - **Data Design and Evaluation:** Designed a unified 1.1M-image dataset with captions, boxes, and masks and used automated model-assisted verification to improve labeling efficiency; won 2nd place in IOD and 4th place in FSOD at CVPR 2025 challenges.
 - **[Lead] ML Training & Serving Platform:** Designed reusable distributed multi-GPU training, evaluation, and deployment infrastructure with PyTorch and AWS Batch; optimized production inference with TensorRT and Triton, improving throughput by 5x over pure PyTorch serving.
 - **Customer Delivery at Scale:** Generated 1,100+ models and delivered 60+ production endpoints across customer tenants in one year.
 - **Post-training and Efficiency:** Built a flexible LoRA, Adapter, and LST post-training pipeline; reduced GPU memory by 65.6% and training time by 44.3% while preserving model performance.
 - **Web ML:** Developed browser, hybrid, and cloud-based interactive segmentation tools using RepViT-SAM, FocalClick, SAM, and SAM2; increased segmentation labeling speed by 25x and built video tracking with fine-tuned personalized SAM.
 - **Transfer Learning:** Developed pretraining and parameter-efficient transfer learning algorithms using DETR architectures and CBNetV2; built multi-domain datasets such as night and aerial imagery with box and polygon annotations.
 - **Uncertainty Estimation:** Implemented Monte Carlo dropout for keypoint and segmentation models, reducing human inspection effort to 30% by ranking predictions with top uncertainty scores.
- **Kakao Enterprise** Pangyo, South Korea
Machine Learning Engineer Intern *Feb 2021 - Aug 2021*
 - **AutoML:** Fixed the scaling issue of AutoGluon's NeuralNetFastAI algorithm and developed a Flask-based training/inference AutoML framework with a simple front-end.

Team ISLAND

Seoul, South Korea

- *Chief Technology Officer*

Mar 2019 - Oct 2020

- **Startup:** Created [ZZAZZ](#), a mobile video-editing platform for composing motion effects around people; led development, investor relations, and product management with a five-person core team. Across leadership roles, led up to eight people simultaneously.
- **Fundraising:** Prepared investor pitch materials and helped raise approximately \$290K through government subsidies and seed funding.
- **Mobile ML:** Developed a lightweight 3D human pose estimation model for mobile devices and converted PyTorch models to TensorFlow Lite.

- **Seerslab Inc.**

Seoul, South Korea

- *Software Engineer Intern*

Jul 2018 - Aug 2018

- **Annotation Tool:** Developed a facial landmark annotation tool using D-lib, Python, and PyQt.

OPEN SOURCE

- **OpenLocalAgent:** Building small tool-calling agents from scratch in pure PyTorch using open data, with pretraining, midtraining, post-training, and evaluation across ten public agent benchmarks. Developed dataset-level loss analysis to investigate differences in predictability across coding, structured, and web-text corpora.
- **OCR:** Building a reproducible evaluation and improvement framework for sub-1B document VLMs, covering accuracy, calibration, robustness, and LoRA adaptation. Unified public document datasets into a shared evaluation and training format, with deduplication, held-out splits, task-specific metrics, and ablation studies.
- **Absorbed:** Developing an open-source SwiftUI roguelike with turn-based combat, branching expeditions, spirit capture, persistent progression, and English/Korean localization. Built deterministic content, asset, localization, and balance checks; currently a playable iOS prototype under active development.
- **Hugging Face Transformers:** Contributed 40+ pull requests to a large production codebase across multimodal and vision models, independently shipping model architectures, processors, conversion scripts, distributed training fixes, tests, and documentation.
- **Hugging Face Transformers - Segment Anything 2:** Led the integration of SAM2 into Transformers across model architecture, image/video processing, conversion scripts, documentation, and integration tests. The PR was merged on 2025-08-13 after a long community review cycle and is credited in the official [SAM2 documentation](#).
- **Hugging Face Transformers - Molmo2:** Opened Molmo2 support for Transformers, including model integration work for a multimodal VLM family, and published [danelcsb/Molmo2-4B](#) on the Hugging Face Hub.
- **RT-DETR:** Implemented RT-DETR support in Transformers.
- **ViTPose:** Implemented ViTPose support in Transformers.
- **Training and Distributed Fixes:** Fixed DETA auxiliary training loss issue and improved gathering behavior in multi-GPU training pipelines; additional contributions are available in [Transformers pull requests](#).

PAPERS

- **ZERO: Multi-modal Prompt-based Visual Grounding:** arXiv - Jul 6, 2025
- **MobileHumanPose: Toward real-time 3D human pose estimation in mobile devices:** IEEE/CVF CVPR Workshop - Jun 19, 2021
- **Improved DOA estimation of MUSIC algorithm based on irregular microphone array:** UIUC Research - Dec 20, 2018
- **On Joint Transfer of Energy and Information: A Markov Decision Problem Formulation:** Research paper - Dec 13, 2018
- **Terahertz Signal Analysis in Tissue of Gastroesophageal Reflux Disease:** Sejong Science High School - Dec 31, 2013

PRESENTATIONS

- **Building Vision Foundation Model in Startup:** Superb AI First Online Seminar - Oct 29, 2025
- **Automation in AutoLabeling:** MLOps World: Machine Learning in Production Bay Area - Jun 19, 2022

EDUCATION

- **KAIST (Korea Advanced Institute of Science and Technology)** Daejeon, South Korea
Professional Master in Electrical Engineering; GPA: 3.55/4.3
Feb. 2020 – Aug. 2021
- **POSTECH (Pohang University of Science and Technology)** Pohang, South Korea
Bachelor in Electrical Engineering; GPA: 3.4/4.3
Mar. 2015 – Feb. 2020
- **UIUC (University of Illinois Urbana-Champaign)** Illinois, United States
Exchange Student; GPA: 3.6/4.0
Jan. 2018 – Dec. 2018
- **Sejong Science High School** Seoul, South Korea
One year early graduation
Mar. 2013 – Feb. 2015

PROGRAMMING SKILLS

- **Research:** Python, PyTorch, Transformers, multimodal pre/post-training, LoRA, contrastive alignment, evaluation
- **Systems:** Linux, AWS Batch, multi-GPU training, MLflow, TensorRT, Triton, ONNX, TensorFlow Lite, Git
- **Agents:** LangChain, Langfuse, Claude Code, Codex, opencode
- **Languages:** Korean (native), English (professional)